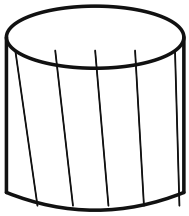
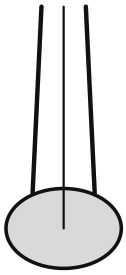
PHYSICAL AND CHEMICAL SCIENCE

Sound

Vibrations, decibels, how sound travels, sound insulation, and pitch in musical instruments



dhol (drum)



ektara (string)



bansuri (flute)

In this pack

- **Learn it:** Sound: vibrations, loudness and pitch ★☆☆
- **Fact check:** What vibrates? ★☆☆
- **Think it through:** Thinking about sound ★★☆
- **Investigate:** Which material blocks sound best? ★★☆
- **Real-world case study:** Too much noise! ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 4 Our five senses (hearing).

This pack belongs to: _____ **Class:** _____

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Sound: vibrations, loudness and pitch

★☆☆ Grade 3

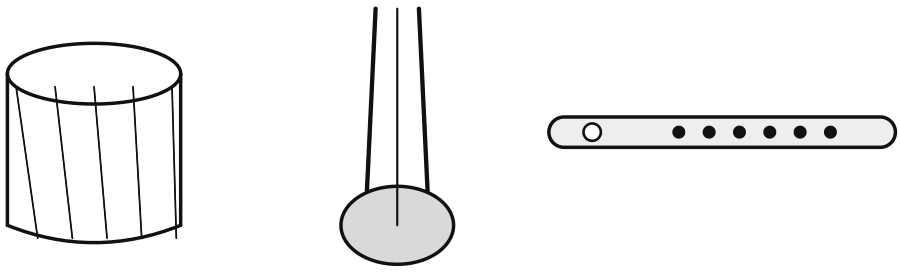
Name: Class: Date:

Learning goals

- Explain that sounds are made by vibrations.
- Measure loudness in decibels and describe how sound travels.
- Explain pitch and how musical instruments change it.

Put your fingers on your throat and hum. Can you feel a tingle? That is a **vibration**: a very fast shaking. **Every sound is made by something vibrating.**

- When you hit a **dhol**, its skin vibrates.
- When you pluck an **ektara**, its string vibrates.
- When you blow a **bansuri**, the air inside it vibrates.
- When you talk, your **voice box** in your throat vibrates.

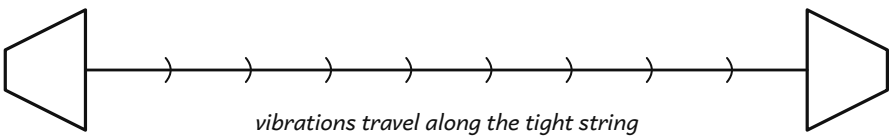


dhol (drum) **ektara (string)** **bansuri (flute)**

Instruments from Bangladesh.

How sound travels

The vibrations pass from one tiny particle to the next until they reach your **ear**. Sound can travel through **gases** (air), **liquids** (water) and **solids** (wood, metal, string). It travels better through solids: put your ear on a desk while a friend taps the other end! Sound cannot travel through empty **space**, where there is no air.

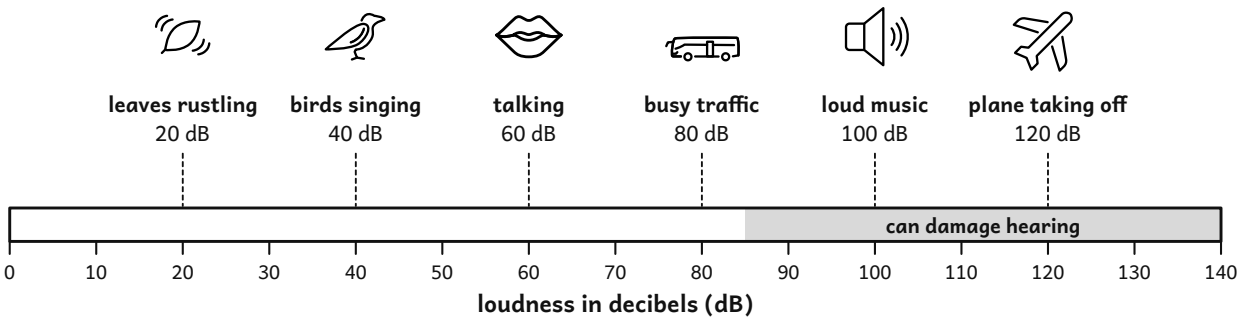


A string telephone.

How loud?

Loudness (volume) is measured in **decibels (dB)** with a **sound level meter**. Many mobile phones have a sound meter app. Sounds louder than about **85 dB** can damage

your hearing if you listen for a long time.



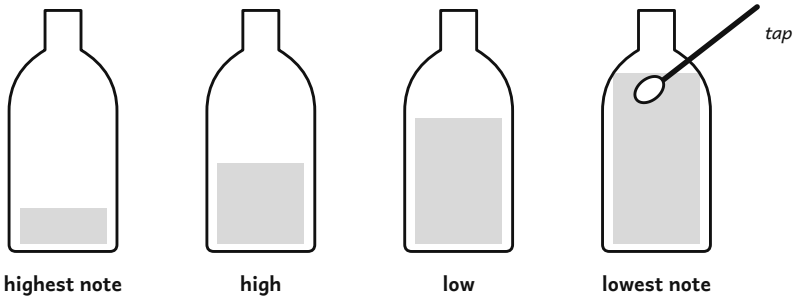
Everyday sounds on the decibel scale.

Soft, thick materials such as **foam**, **cloth**, **curtains** and **egg boxes** soak up sound. They stop it from travelling through. This is called **sound insulation**. Ear defenders use it to protect workers' ears.

High and low: pitch

Pitch is how **high** or **low** a sound is. A bird tweets with a high pitch; a big drum booms with a low pitch. High and low sounds can each be loud or soft.

Higher pitch	Lower pitch
short, thin or tight strings	long, thick or loose strings
small drums, tight drum skins	big drums, loose drum skins
short tube of air (cover fewer holes on a flute)	long tube of air (cover more holes)



Tap bottles with different amounts of water.

Key words

vibrate	shake very fast
volume	how loud a sound is
decibel (dB)	the unit for measuring loudness
pitch	how high or low a sound is
sound insulation	a material that stops sound travelling through

FACT CHECK · C2 · SOUND

What vibrates?

★☆☆ Grade 3

Name: Class: Date:

Learning goals

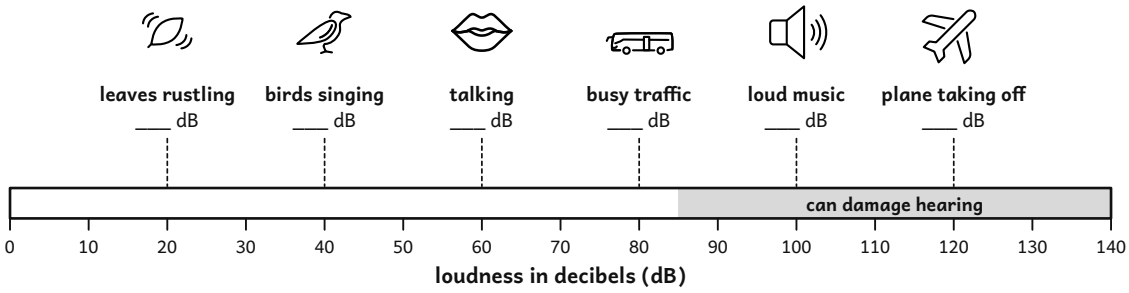
- Name what vibrates.
- Read the decibel scale.

1 Draw a line to match the instrument to the part that vibrates. [4]

Draw a line to match each pair, or write the letter.

dhol ●	● **A** the drum skin
ektara ●	● **B** the air inside
bansuri ●	● **C** the string
your voice ●	● **D** the voice box in your throat

2 Write the loudness of each sound. [3]



leaves	birds	talking	traffic	loud music	plane

3 True or false? [3]

Statement	True	False	If false, correct it
1. Sound can travel through water.			
2. Sound travels through space.			
3. A loud sound always has a high pitch.			

Thinking about sound

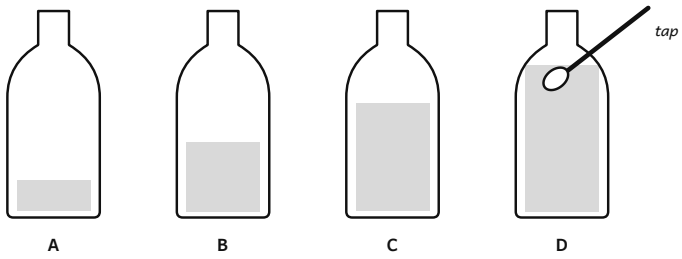
★★☆
Grade 3

Name: Class: Date:

Learning goals

- Explain pitch and loudness.

1 **Which bottle makes the highest note when tapped? Which makes the lowest?** [2]



2 **An ektara player presses the string to make it shorter. What happens to the pitch?** [1]

3 **Why do people who work with loud machines wear ear defenders?** [2]

4 **A string phone only works when the string is tight. Why?** [2]

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LEARN · PLAY · WONDER

Which material blocks sound best?

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Name: Class: Date:

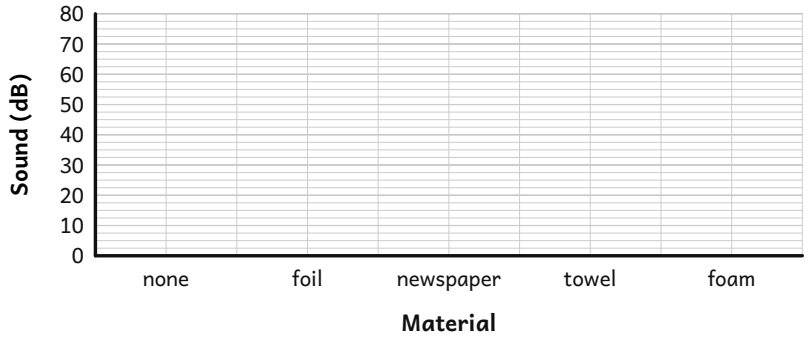
Learning goals

- Measure sound in decibels.
- Draw a bar chart and conclude.

Class 3C put a ringing alarm clock inside a box and wrapped it in different materials. They measured the sound 1 metre away with a sound meter app.

1 **Draw a bar chart.** [3]

Material	none	foil	newspaper	towel	foam
Sound (dB)	70	66	60	50	45



2 **Which material blocked the sound best?** [1]

.....

3 **What kind of materials are good sound insulators?** [2]

.....

.....

4 **What did they keep the same to make it fair?** [1]

.....

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REAL-WORLD CASE STUDY · C2 · SOUND

Too much noise!

★★★
Grade 3

Name: **Class:** **Date:**

Learning goals

- Use what you know about sound to solve a real problem.

Science | Health | Citizenship

Dhaka is one of the noisiest cities in the world. On busy roads the sound is often above **85 dB** because of car horns, buses, building work and loudspeakers. Traffic police stand in this noise all day.

Too much noise can damage hearing and make it hard to sleep, learn and think. Bangladesh has rules: very loud hydraulic horns are not allowed, and roads near **schools and hospitals** are **silent zones** where drivers must not honk.

Based on reports about noise pollution in Dhaka and Bangladesh's noise rules.

1 **Name three things that make Dhaka noisy.** [3]

2 **Why are roads near hospitals and schools silent zones?** [2]

3 **Make a "Do not honk" sign for your school gate.** [2]

Silent zone sign

REVIEW AND CHALLENGE · C2 · SOUND

Show what you know

★★★
Grade 3

Name: **Class:** **Date:**

Learning goals

- Use sound words.

1 Find the words.

J	U	L	O	U	D	V	Y	P
E	S	L	W	V	T	V	G	I
H	A	U	Y	I	W	K	D	T
D	I	R	U	B	T	G	E	C
N	R	L	G	R	I	N	C	H
O	S	U	R	A	M	B	I	I
I	V	O	M	T	H	W	B	U
S	K	G	W	E	B	B	E	K
E	N	J	B	V	D	O	L	H

decibel drum ear loud noise pitch vibrate

2 Loudness is measured in... [1]

◯ metres ◯ decibels

◯ grams

3 A short, tight string makes a... [1]

◯ high pitch ◯ low pitch

◯ no sound

4 Challenge. Design a quiet corner for your classroom. Which materials would you use? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain that sounds come from vibrations	○	○	○
measure sound in decibels	○	○	○
say which materials block sound	○	○	○
explain pitch	○	○	○
change the pitch of an instrument	○	○	○